

AYUSH GANDHI

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EDUCATION

University of Waterloo

Waterloo, ON

Bachelor of Applied Science in Honors Mechatronics Engineering (Co-op program) | 2025 – 2030

- Awards: President's Scholarship of Distinction
- Relevant Courses: Digital Computation, Mechatronics Engineering Designing, Data Structures and Algorithms

EXPERIENCE

Humanoid Team Member

May 2026 – Present

WATonomous / University of Waterloo

- Contributing to the mechanical design of a **15-DOF** humanoid robotic arm using **Onshape** by assisting with forearm/joint design, prototyping, design review and component assembly to ensure dexterous hand mobility.
- Collaborating with multidisciplinary robotics teas to iterate designs for manufacturability and system assembly.

Engineering and Trades Learning Assistant

Jan. 2026 – Apr. 2026

Sheridan College | Brampton, ON

- Ran **CAD**, **electronics**, and fabrication sessions, helping students debug wiring, fix **driver** configs, and get hardware working.
- Caught design problems early by reviewing CAD files for **DFM** and **GD&T** issues before anything got printed or machined.
- Covered hands-on use of **milling machines**, **lathes**, and 3D printers with students building original mechanical projects.

Mechanical Team Lead

Sep. 2022 – Jun. 2023

F.I.R.S.T. Robotics / École Secondaire Pierre-Laporte

- Ran the **20-person** mechanical build team through a full competitive season and hit every single deadline.
- Designed an **800 g** motorized arm in **SolidWorks**, ran load analysis on the model, and got **15%** better mechanical efficiency out of the final version.

PROJECTS

6 D.O.F. Motorized Robotic Arm

Mar. 2026 – Present

Personal Project | Arduino, PCA9685, A4988, OpenCV, Python, MediaPipe

- Designed every joint in **Onshape** and printed the structure on an **FDM printer**, landing at a **15 cm** reach and **150 g** payload.
- Wired up a **PCA9685** PWM board and **A4988** stepper drivers to an **Arduino** to control all 6 joints independently over **I2C**.
- Building out **gesture-based** control using a webcam feed, **MediaPipe** hand tracking, and a custom **IK** solver written in **Python**.

Mars Explorer: Omnidirectional Rover

Ongoing

Personal Project | ESP32, Mecanum Wheels, Ultrasonic Sensors, Python, Web UI

- Designed the drivetrain around **Mecanum** wheels so the rover can strafe, spin, and translate in any direction without turning.
- Built a rotating ultrasonic mount that continuously sweeps and builds a live obstacle map of the surrounding area.
- Wrote a **browser-based UI** from scratch so you can drive it manually with a **joystick** and watch **telemetry** at the same time.

Autonomous Object Sorting Robot

Oct. 2026 – Dec. 2026

University of Waterloo | C++, Python, PID Control, Sensor Fusion

- Reworked the **sensor fusion** weighting in the classification pipeline and pushed accuracy from **60% up to 90%**.
- Wrote **PID** loops for both drive and steering so the robot reliably hits its waypoints and routes itself to the shortest path.
- Fit all the hardware into a tight chassis without giving up sensor coverage or drive performance.

Saturn-V Multi-Stage Flight Simulator

Mar. 2025 – May 2025

Dawson College | Python, NumPy, Matplotlib

- Coded a multi-stage rocket simulator in Python covering fuel burn rates, thrust curves, and stage-separation mechanics.
- Pulled **NASA** telemetry data to validate the model and kept iterating on the physics until outputs were within **5%** of data.

TECHNICAL SKILLS

Software and tools: C++, Python, OpenCV, MediaPipe, JS, Tailwind CSS, HTML, PyTorch, NumPy, Matplotlib, MS Office

Hardware and Embedded: Arduino, ESP32, PCA9685, A4988, Ultrasonic Sensors, Motor Drivers, I2C

CAD/Production: SolidWorks, Onshape, Fusion 360, AutoCAD, 3D Printing, Milling, Lathe, Rapid Prototyping, DFM, GD&T

Languages: English, French, Gujarati, Hindi